Fig. 4: Testing set of cat and dog images not included in training set

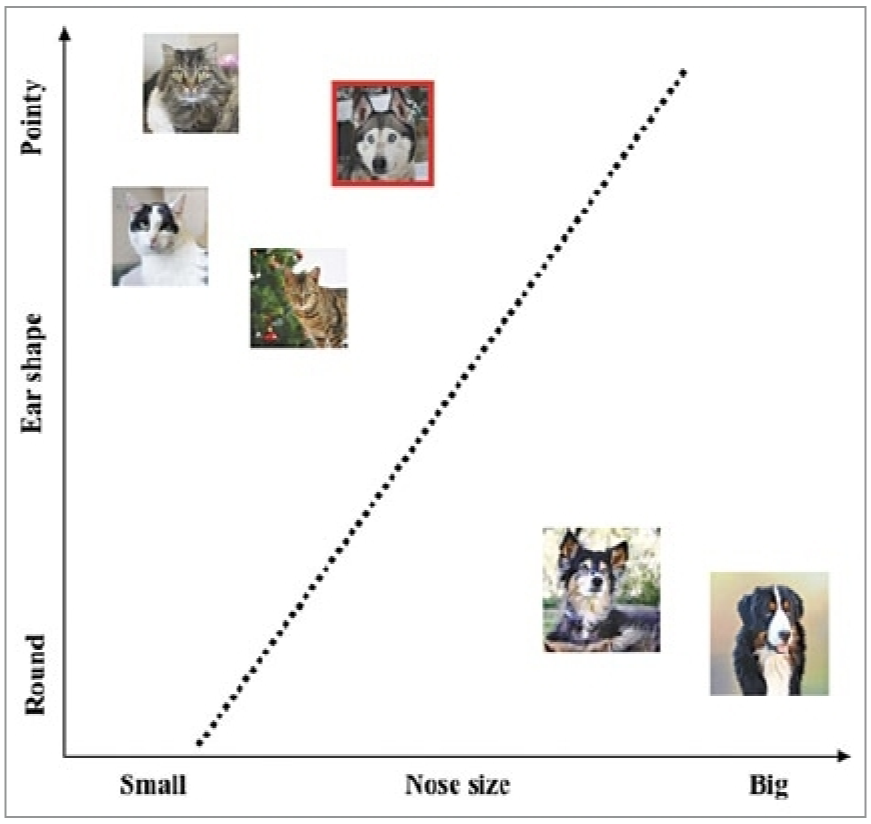

Fig. 5: Classification of test images using trained linear model

For example, you can easily extract two features from each image of the training set, that is, size of the nose (relative to size of the head—from small to big) and shape of the ears (from round to pointed). By examining the images in the training set (Fig. 1), you can observe that all cats have a small nose and pointed ears, while dogs have a long nose and round ears.

With the current choice of features, each image can be represented using a number expressing relative nose size and another number capturing pointed or round ears. Hence, you can represent our training set in a two-dimensional feature space where the feature nose size and ears shape are on horizontal and vertical axes, respectively, as shown in Fig. 2.

Training. After you have a good feature representation of the images in the training set, the final task is to teach a computer how to distinguish between cats and dogs. It is a simple geometric problem where the computer must find a line that separates the two images in the designed feature space.

Since the line is parameterised by a slope and intercept, finding the right values for both is important. Fig. 3 shows the trained line that divides the feature space into cats and dogs, respectively. Once the line is obtained, any image added later—whose features lie above the line—is considered as cat by the computer. Any image that falls below the line is considered as dog.

Model testing. To test the computer on how well it has learned to distinguish between cats and dogs, provide it with a series of images not seen earlier and see how well it can identify the animal in each image. In Fig. 4, a sample of images (of new cats and dogs), not seen by the computer during training phase, is shown.

What the computer does is, it takes the features into account and checks which side of the line these images fall. For instance, it is observed in Fig. 5 that all new cats and dogs (except one) from the testing set were identified accurately. Misidentification of one dog (shown in red box) is due to the choice of features that were designed using the training set, as shown in Fig. 1.

This dog was identified as a cat because size of its nose and shape of its ears match that of cats in the training set. To avoid this issue, train the computer with more data and use more discriminating features that will help distinguish cats and dogs better.

To supplement this article, Python script for recognising cats from other animals is provided. It contains a database that has all kinds of animals in a folder. The algorithm predicts whether the image is a cat or not.

Testing procedure

1. Install Python version 3.6.6 on Windows 8.1 operating system.

2. Install required libraries listed in main_program.py code, including pillow, numpy, matplotlib, scipy, and h5py. All relevant files and images should be kept under the same folder as main_program.py file.

3. Run main_program.py from the command line or IDLE (GUI) from the desktop.

4. Type in the image file name, say, test2.jpg. This is the image file name of a cat in this example. The computer displays the output as 'This is a cat' on the terminal. If the image is not of a cat, it displays 'This is not a cat.' EFY

EFY Note

The source code of this project is available for free download at *source.efymag.com*

Dr Ahlad Kumar is currently doing research in Concordia University, Montreal, Canada in the field of image processing. He is interested in the field of image restoration